Homotopy.

Def. Let $f, g: X \to Y$ be continuous maps.
We said that f and g are Homotopic if: there exists a Continuous map
$F: X \times I \to Y$ such that $\begin{cases} F(x,0) = f(x), & \forall x \in X. \\ F(x,1) = g(x), & \forall x \in X \end{cases}$
This F is called a Homotopy and in such a case, denote $f \simeq g$.

Thm. Let $C(X,Y) \overset{\text{def}}{=} \{f: X \to Y \mid f \text{ is continuous}\}$,
Define a relation on $C(X,Y)$ as: $f \sim g \Leftrightarrow f \simeq g$.
Then, the Homotopic is equivalent relation.
Proof. [Reflexivity: given $f \in C(X,Y)$, define

$$F: X \times I \to Y: (x,t) \mapsto f(x)$$

Then, since $F^{-1}[V] = f^{-1}[V] \times I$, F is continuous such that $F(x,0) = F(x,1) = f(x)$.
[Symmetric. Let $f \simeq g$ via $F: X \times I \to Y$. Define $G: X \times I \to Y: (x,t) \mapsto F(x,1-t)$
Then, $G(x,0) = F(x,1) = g(x)$ and $G(x,1) = F(x,0) = f(x)$.
[Transivity. Let $f \simeq g$ via F and $g \simeq h$ via G. Define

$$H: X \times I \to Y: (x,t) \mapsto \begin{cases} F(x,2t) & \text{if } 0 \le t \le \frac{1}{2} \\ G(x,2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases}$$

Then, since $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ are closed with $F(x,1) = g(x) = G(x,0)$, therefore, the Pasting lemma gives that the H is Homotopy from f to h. So $f \simeq h$.

Def. Let X and Y be topological space.
We said that X and Y are Homotopy equivalent if:
there exises two Continuous map $f: X \to Y$ and $g: Y \to X$ such that

$$g \circ f \simeq id_X \quad \text{and} \quad f \circ g \simeq id_Y.$$

In such a case, denote $X \simeq Y$.
And, such a $f: X \to Y$ is called a Homotopy equivalence,
and such a $g: Y \to X$ is called a Homotopy inverse.

Note: Trivially, Homeomorphism is Homotopy equivalence. Therefore,

$$X \cong Y \Rightarrow X \simeq Y,$$

but the Converse does not hold.
Counterexample: Consider two subspaces $S^1, \mathbb{R}^2 \setminus \{0\} \subseteq \mathbb{R}^2$.
First, since S^1 is closed and bounded, so it is Compact.
And, since $\{0\}$ is closed, $\mathbb{R}^2 \setminus \{0\}$ is open, so it is not Compact. So, $S^1 \not\cong \mathbb{R}^2 \setminus \{0\}$
Next, to show that $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$, Consider that an inclusion

$$f: S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\} : x \mapsto x$$

and

$$g: \mathbb{R}^2 \setminus \{0\} \to S^1 : x \mapsto \frac{x}{\|x\|}.$$

Then, both f and g are Continuous, and $g \circ f = id_{S^1}$.
And, $(f \circ g): \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\} : x \mapsto x/\|x\|$ is Homotopic to $id_{\mathbb{R}^2 \setminus \{0\}}$ via

$$F: [\mathbb{R}^2 \setminus \{0\}] \times I \to \mathbb{R}^2 \setminus \{0\} : (x, t) \mapsto (1-t) \cdot \frac{x}{\|x\|} + t \cdot x.$$

which satisfies $F(x, 0) = x/\|x\|$ and $F(x, 1) = x$, so Prove.

Contractible

Def.
A space that homotopy equivalent to a point is called Contractible.

Thm. Let X be a Topological space. Then, TFAE:
a). X is Contractible. (Homotopy equivalent to a point)
b). $id_X : X \to X ; x \mapsto x$ is null-homotopic.
c). Given $x \in X$, $C_x \simeq id_X$ where $C_x : X \to X ; z \mapsto x$.

Proof.
Assume a). That is, there exist two maps
$f : X \to \{p\}$ and $g : \{p\} \to X$ such that $f \circ g \simeq id_p$ and $g \circ f \simeq id_X$.

Then, since $f : X \to \{p\}$ must be a constant mapping $x \mapsto p$,
$g \circ f(x) = g(p)$ for all $x \in X$, so $g \circ f$ is constant, so id_X is null-homotopic.
It proved b).
Assume b). Let $C : X \to X ; z \to a$ be a constant map such that $C \simeq id_X$, via
$H : X \to I \to X$ such that $H(z,0) = z$ and $H(z,1) = a$.

Let $x \in X$ be fixed. Define a path $P : [0,1] \to X ; t \mapsto H(x,t)$.
Then, $P(0) = H(x,0) = x$ and $P(1) = H(x,1) = a$.
Now, $F : X \times I \to X ; (x,t) \mapsto P(t)$ is the Homotopy to $C \simeq C_x$. Now, by transitivity.
$C \simeq id_X$ and $C \simeq C_x$ implies that $C_x \simeq id_X$. It proved c).
Assume c). Let $f : X \to \{p\} ; x \mapsto p$ be a map and let $x \in X$ be given.
Then, a map $g : \{p\} \to X ; p \mapsto x$ satisfies

$$f \circ g = id_p \text{ and } g \circ f = C_x.$$

By c), $C_x \simeq id_X$, so $g \circ f \simeq id_X$. So proved ∎

Corollary. Contractible space is Path-Connected.
Proof. Suppose that X is Contractible space. Let $x, y \in X$ be given.
Then, $C_x \simeq id_X \simeq C_y$, so there exists a Homotopy

$$F : X \times I \to X \text{ such that } \begin{cases} F(z,0) = C_x(z) = x & \text{for all } z \in X \\ F(z,1) = C_y(z) = y & \text{for all } z \in X. \end{cases}$$

Now, $P : I \to X ; t \mapsto F(x,t)$ is a path from x to y. ∎

[Basic propositions,

[Prop. Let $f, g: X \to Y$ be continuous and $f \simeq g$.
If f is Homotopy equivalence, then g is also Homotopy equivalence

Proof. Let $h: Y \to X$ be the Homotopy inverse of f.

$h \circ f \simeq id_X$ and $f \circ h \simeq id_Y$.

Meanwhile, let $F: X \times I \to Y$ be the Homotopy from f to g.
Then, $H: X \times I \to X : (x,t) \mapsto h(F(x,t))$ satisfying $\begin{cases} H(x,0) = h(f(x)) \\ H(x,1) = h(g(x)) \end{cases}$

So, $h \circ f \simeq h \circ g$, thus $h \circ g \simeq id_X$. And, define

$G: Y \times I \to Y : (y,t) \mapsto F(h(y),t)$ satisfying $\begin{cases} G(y,0) = f(h(y)) \\ G(y,1) = g(h(y)) \end{cases}$

So $f \circ h \simeq g \circ h$, thus $g \circ h \simeq id_Y$. So proved. ☑

[Prop. Let $f, g: X \to \prod_{\alpha \in \Lambda} Y_\alpha$ be Continuous maps. Then, for projection π_α,

$f \simeq g$ if and only if for all $\alpha \in \Lambda$, $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$.

Proof. Suppose that there is a Homotopy $F: X \times I \to \prod Y_\alpha$. (That is, $f \simeq g$).
Then, the projection $\pi_\beta: \prod Y_\alpha$ induces a Homotopy

$(\pi_\beta \circ F): X \times I \to Y_\beta : (x,t) \mapsto \pi_\beta(F(x,t))$.

Satisfying $(\pi_\beta \circ F)(x,0) = (\pi_\beta \circ f)(x)$ and $(\pi_\beta \circ F)(x,1) = (\pi_\beta \circ g)(x)$, so that $\pi_\beta \circ f \simeq \pi_\beta \circ g$. Conversely, suppose that for all $\alpha \in \Lambda$, $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$.
For each $\alpha \in \Lambda$, let $F_\alpha: X \times I \to Y_\alpha$ be the Homotopy from $\pi_\alpha \circ f$ to $\pi_\alpha \circ g$
Define $F: X \times I \to \prod Y_\alpha : (x,t) \mapsto (F_\alpha(x,t))_{\alpha \in \Lambda}$

Then, by the property of product space, F is continuous, so,
it is a Homotopy from f to g.

[Prop. If $X_\alpha \simeq Y_\alpha$ for all $\alpha \in \Lambda$, then $\prod_{\alpha \in \Lambda} X_\alpha \simeq \prod_{\alpha \in \Lambda} Y_\alpha$.

Proof.

[Prop. If Y is Contractible, then $X \times Y \simeq X$.

Proof. Let $P = \{p\}$ be a one-point set space which is Homotopy equivalent to Y. That is, there exist two Continuous maps $f: P \to Y$ and $g: Y \to P$ satisfy

$$f \circ g \simeq id_Y \quad \text{and} \quad g \circ f \simeq id_P.$$

Actually, $g \circ f$ must equals to id_P. To show $X \times Y \simeq X \times P$, define the Homotopy equivalence

$$(id_X \times g): X \times Y \to X \times P : (x, y) \mapsto (x, p)$$

with the Homotopy inverse

$$(id_X \times f): X \times P \to X \times Y : (x, p) \mapsto (x, f(p)).$$

Then, $(id_X \times g) \circ (id_X \times f) \equiv id_X \times id_P = id_{X \times P}$.

Meanwhile, to show $(id_X \times f) \circ (id_X \times g) \simeq id_{X \times Y}$, let

$$F: Y \times I \to Y \quad \text{such that} \quad F(y, 0) = (f \circ g)(y) \text{ and } F(y, 1) \equiv id_Y.$$

be the Homotopy. Now, the map

$$H: (X \times Y) \times I \to X \times Y : ((x, y), t) \mapsto (x, F(y, t))$$

Finally, $H(\cdot, 0) = (x, (f \circ g)(y))$ and $H(\cdot, 1) = (x, id_Y(y)) = (x, y)$, so $(id_X \times f) \circ (id_X \times g) \simeq id_{X \times Y}$. Consequently, $X \times Y \simeq X \times \{p\}$ and since $X \cong X \times \{p\}$ thus $X \times Y \simeq X \times \{p\} \simeq X$. ∎

[Prop. Let X be any space, and Y be a Contractible.
Then, for any continuous map $f, g: X \to Y$, $f \simeq g$.
Moreover, every continuous map $h: X \to Y$ is null-homotopic.

[Prop. Let X be a Topological space, and let

$$CX = X \times I / X \times \{1\}$$

be a Cone, where $I = [0,1]$.

Then, CX is contractible. Moreover, for any space Y and continuous $f: X \to Y$, $f: X \to Y$ is null-Homotopic if and only if there is an extension $\tilde{f}: CX \to Y$.

Proof. Let $P = \{P_0\}$ for fixed $P_0 = [(x_0, 1)] \in CX$.

$f: CX \to P ; x \mapsto P_0$ is a Constant map, so continuous with homotopy inverse $g: P \to CX ; P_0 \mapsto P_0$ So that $f \circ g = id_P \simeq id_P$.

And, $g \circ f(x) = P_0$ is a Constant map, let's show $g \circ f \simeq id_{CX}$.

Construct a Homotopy between $g \circ f$ and id_{CX}. Define

$$H: CX \times I \to CX ; ([(x,s)], t) \mapsto [(x, (1-t) \cdot s + t)]$$

Then, $H([x,s], 0) = [(x,s)]$ and $H([(x,s)], 1) = [(x,1)]$

(Specifically, H is well-defined Continuous map,

(Want: $H([(x,s)], 0) = [($